



Daily Grind Tracker

Complete Technical Project Documentation

⚡ Node.js + Express

🌿 MongoDB Atlas

🍷 Vanilla JS SPA

🚀 Render.com

Project: Daily Grind Tracker (751-Day Quest)

GitHub: <https://github.com/UrQtSketch/Daily-Grind-Tracker>

Live URL: <https://daily-grind-tracker.onrender.com>

Version: v2.8.1 · **Date:** September 17, 2026

Author / Owner: Sketch (UrQtSketch)

Support Email: support.dailygrind@gmail.com



Table of Contents

1	Project Overview & Purpose	
2	Tech Stack & Architecture	
3	File Structure & Codebase	
4	Database Design (MongoDB Atlas)	
5	All API Endpoints	
6	Frontend Features	
7	Authentication System	
8	Real-Time Live Leaderboard (SSE)	
9	Admin Command Center	
10	Security & DDoS Protection	
11	Deployment & Hosting	
12	Load Test Results	
13	Key Credentials & Config	
14	Git History & Local Setup	



1. Project Overview & Purpose

SECTION 01

Daily Grind Tracker is a **full-stack productivity web application** built around a 751-day personal discipline challenge. Users register with Gmail, log daily tasks, earn trophies for milestones, compete on a live leaderboard, and take skill quizzes.



CORE MISSION

751-Day Quest

Build and maintain daily discipline habits over 751 consecutive days



TARGET USERS

Grinders & Hustlers

Students, freelancers, fitness enthusiasts committed to self-improvement



GAMIFICATION

Trophy System

Earn trophies for streaks, quiz wins, milestones. Personal Trophy Vault.



REAL-TIME

Live Leaderboard

SSE pushes instant rank updates to all connected users

Core Features



Daily Task Tracker

Log tasks, mark completion, pick dates. Progress auto-saves. 7-day inactivity triggers penalty modal.



Trophy Vault & Celebrations

Earn trophies at Day 1/7/30/100/300/500/751 + Quiz wins + Streaks. Full-screen confetti + optional YouTube reward song.



Skill Quiz Arena

Multi-category quizzes (Coding, GK, Math, Science, History, Sports). 20 questions with timer. Trophy on win.



Live Leaderboard

Top 10 + All Users ranked by trophies. Real-time via SSE. Updates instantly when any user earns a trophy.



2. Tech Stack & Architecture

SECTION 02



RUNTIME

Node.js v18+

CommonJS · Single-threaded event loop



FRAMEWORK

Express v5

REST API · Static files · Middleware



DATABASE

MongoDB Atlas

Cloud NoSQL · Mongoose ODM · JSON fallback



FRONTEND

Vanilla JS SPA

No framework · Dynamic HTML rendering



STYLING

Pure CSS3

Glassmorphism · CSS Grid/Flex · Dark theme



REAL-TIME

SSE

Native EventSource · No WebSocket needed



EMAIL

Nodemailer

OTP + daily reminders via Gmail SMTP



AUTH

Custom Sessions

crypto.randomUUID() · Bearer token

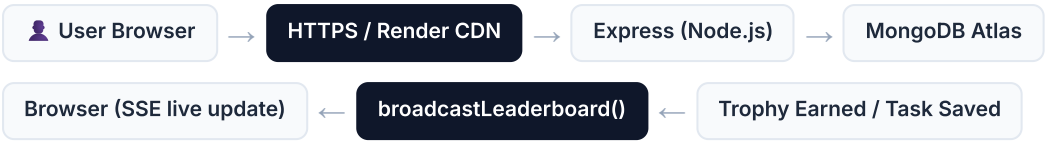


SECURITY

Helmet + RateLimit

HTTP headers · 3-layer rate limiting

Architecture Flow



npm Dependencies

Package	Version	Purpose
express	^5.2.1	Web framework — REST API server, routing, static files
mongoose	^9.10.1	MongoDB ODM — schema, queries, connection management
cors	^2.8.6	Cross-Origin Resource Sharing middleware
nodemailer	^10.0.10	Email — OTP verification & daily reminders
express-rate-limit	^8.7.0	Rate limiting — DDoS & brute-force protection
helmet	^8.3.0	Security HTTP headers — XSS, clickjacking prevention

3. File Structure & Codebase

SECTION 03

```
Daily-Grind-Tracker/
├─ server.js           (20 KB)  - Express server, all API routes, SSE stream, security middleware
├─ database.js         (26 KB)  - MongoDB + JSON fallback DB layer, all queries
├─ app.js              (168 KB) - Frontend SPA logic, UI rendering, all 7 views
├─ quiz.js             (157 KB) - Quiz Arena engine, 500+ questions, scoring system
├─ auth.js             (7 KB)   - Client-side auth: login/register/logout/token
├─ styles.css          (128 KB) - All CSS: glassmorphism, dark theme, animations
├─ index.html          (23 KB)  - Main SPA shell: sidebar, topbar, modals
├─ login.html          (6 KB)   - Login page UI
├─ register.html       (8 KB)   - Register + OTP verification page
├─ mailer.js           (16 KB)  - HTML email templates, SMTP configuration
├─ reminderService.js  (5 KB)   - IST timezone scheduler, 8PM daily reminders
├─ package.json         - Project metadata, scripts, dependencies
├─ assets/             - Static images, icons
├─ data/               - Local JSON fallback storage
│   └─ tracker_db.json  - Users, sessions, tracker state
└─ support_tickets.json - Support messages
```

File Responsibilities

File	Type	Lines	Key Responsibility
server.js	Backend	610	All REST API endpoints, SSE stream, middleware, auth guards
database.js	Backend	~800	MongoDB schemas, CRUD operations, leaderboard, ban system
app.js	Frontend	3,289	Entire frontend SPA — all views rendered dynamically
quiz.js	Frontend	~4,000	Quiz engine, 500+ questions, 6 categories, timer, scoring
auth.js	Frontend	~200	Client auth wrapper: getToken(), current(), logout()
styles.css	CSS	2,860+	Full dark glassmorphism theme, responsive design, all components
mailer.js	Backend	~500	HTML email templates for OTP and daily reminder emails

4. Database Design — MongoDB Atlas

SECTION 04



Dual-Mode Storage
— MONGODB_URI set → uses MongoDB Atlas (cloud). Not set → falls back to local /data/ JSON files automatically.

MongoDB Collections

Collection	Key Fields	Purpose
users	email, name, password (hashed), trophies, trackerState, createdAt, lastActiveDate, isBanned	All user accounts and progress data
sessions	token (UUID), email, userAgent, createdAt, expiresAt	Active login sessions — token-based auth
otps	email, name, password, otp, attempts, expiresAt	Pending OTP verifications (TTL: 10 min, max 5 attempts)
bans	email, reason, bannedAt, bannedBy	Permanently banned Gmail addresses
support_tickets	name, email, category, message, createdAt	User support messages — visible in Admin panel

Tracker State Schema (stored inside user document)

```
{
  "tasks":      [ { "id":"uuid", "text":"Task name", "done":true, "date":"2026-09-17" } ],
  "trophies":   5,
  "trophyLog":  [ { "name":"Day 7 Warrior", "icon":"🏆", "earnedAt":"2026-09-10" } ],
  "currentStreak": 12, "bestStreak": 25,
  "startDate":  "2026-08-01", "dayNumber": 47,
  "lastActiveDate":"2026-09-17",
  "skills":     { "coding": 3, "fitness": 5, "reading": 2 }
}
```



5. All API Endpoints

SECTION 05

Authentication Routes

POST	/api/auth/send-otp	Send 6-digit OTP to Gmail. Validates name, email (@gmail only), password.
POST	/api/auth/verify-otp	Verify OTP → creates user account + session token. Max 5 attempts.
POST	/api/auth/resend-otp	Resend new OTP to same email, replaces old OTP.
POST	/api/auth/login	Login with email + password → returns session token. Rate limited: 20/15min.
POST	/api/auth/logout	Invalidates session token. Requires Bearer auth header.
GET	/api/auth/me	Returns current authenticated user profile.

Tracker & Leaderboard

GET	/api/tracker	Load user's tracker state (tasks, trophies, streaks). Auth required.
PUT	/api/tracker	Save tracker state → triggers live leaderboard broadcast. Auth required.
GET	/api/leaderboard	Snapshot: Top 10 + All Users with ranks. Optional auth for personal rank.
GET	/api/leaderboard/stream	SSE stream — persistent connection, server pushes live updates via EventSource.

Admin Routes (x-admin-key header required)

GET	/api/admin/check	Check admin access. Returns isAdmin boolean.
GET	/api/admin/users	Full registered user list with status, activity, trophies + support messages.
GET	/api/admin/messages	All support ticket messages from users.
POST	/api/admin/ban	Permanently ban or unban user by email. Terminates all their sessions.

Support & Utility

POST	/api/support/feedback	Submit support message → saved to DB → appears in admin panel.
GET	/api/health	Health check — status, DB type, timestamp. Rate limit skipped.

GET /api/reminders/status

SMTP config status, IST time, reminder mode.

POST /api/reminders/check-now

Force-trigger reminder check. Can bypass 8PM time window.



6. Frontend Features

SECTION 06

The entire frontend is a **Single Page Application (SPA)**. Only one HTML page (index.html). All views rendered dynamically by app.js. Navigation via URL hash (#dashboard, #leaderboard, etc.).

All Views / Pages

View	Hash Route	What It Shows
Dashboard	#dashboard	Hero section, today's tasks checklist, streak counter, day number, task search
Trophy Vault	#vault	All earned trophies with earn dates, trophy titles, progress to next
Skill Arena	#skills	6 skill categories, level tracking, Earn Trophy button → opens Quiz
Leaderboard	#leaderboard	Live Top 10 + All Users ranked, search, LIVE badge, personal rank
Admin Center	#admin	PIN-locked: user management, ban/unban, support inbox
Support & Help	#support	Contact form with category → message sent to admin panel
Settings	#settings	Profile info, notification preferences, account management

DESKTOP (≥768PX)

Sidebar visible by default. : button collapses it. Smooth CSS transition animation.

MOBILE (<768PX)

Sidebar hidden. Menu button opens as overlay with backdrop. Auto-closes on nav click.



7. Authentication System

SECTION 07

Registration Flow (OTP-based)



Security Rules

Rule	Implementation
Only @gmail.com allowed	isGmail() validates server-side — non-Gmail returns 400 error
Password hashing	crypto module — SHA-256 hashing stored in DB, never plain text
Session tokens	crypto.randomUUID() — 128-bit random UUID stored in sessions collection
OTP expiry	10-minute TTL, max 5 wrong attempts before auto-delete
Banned users	Every API request checks ban list — banned Gmail → 403 instantly
Login rate limit	Max 20 login attempts per IP per 15 minutes
Session invalidation	Logout deletes session from DB → token invalid immediately



8. Real-Time Live Leaderboard (SSE)

SECTION 08



Server-Sent Events (SSE)

— Native browser EventSource API. Persistent HTTP connection. Server pushes data to client when trophy is earned or tracker saved. No WebSocket library needed.

1

User Opens Leaderboard

Browser opens persistent connection to `/api/leaderboard/stream`. Server adds to `sseLeaderboardClients` Set.

2

Any User Saves Tracker / Earns Trophy

`PUT /api/tracker` → server calls `broadcastLeaderboardUpdate()` → fresh DB query.

3

Server Pushes to All Connected Clients

Iterates all open SSE connections, writes JSON: `data: {top10, allRanks, eventNotice}`

4

Browser Updates UI Instantly

EventSource listener receives data → re-renders leaderboard without page refresh. LIVE badge pulses.

```
// Keep-alive every 20 seconds (prevents connection timeout)
setInterval(() => {
  for (const client of sseLeaderboardClients) {
    client.write(": keep-alive

");
  }
}, 20000);
```



9. Admin Command Center

SECTION 09

Access Control

Method	Value	Access Level
Master PIN 1	grind751	Full Admin Access
Master PIN 2	admin2026	Full Admin Access
HTTP Header	x-admin-key: grind751	API Admin Access
Owner Emails	deepak222@gmail.com · support.dailygrind@gmail.com	Server-level admin

USER DIRECTORY TAB

All users: Name, Email, Join Date, Last Active, Status, Trophies, Ban/Unban. Search by name or email.

SUPPORT MESSAGES TAB

All messages from Support section. Sender name, email, category, message, date. Gmail reply button.

BAN / UNBAN SYSTEM

Ban → POST /api/admin/ban → permanent Gmail block. All sessions terminated. 403 on next login.

STATS DASHBOARD

Total Registered, Active Today, Banned Users, Support Messages — 4 live stat cards.



10. Security & DDoS Protection

SECTION 10

3-Layer Rate Limiting (express-rate-limit v8)



Global Limiter — All Routes

300 requests / 10 min / IP. Blocks DDoS floods. Health endpoint excluded.

✓ ACTIVE



Auth Limiter — /api/login + /api/register

20 attempts / 15 min / IP. Prevents brute-force password attacks.

✓ ACTIVE



Admin Limiter — /api/admin/*

60 requests / 5 min / IP. Protects admin endpoints from scraping.

✓ ACTIVE

HTTP Security Headers (Helmet.js v8)

Header	Protects Against
X-Content-Type-Options: nosniff	MIME type sniffing attacks
X-Frame-Options: DENY	Clickjacking (iFrame embedding)
X-XSS-Protection	Reflected XSS in older browsers
Strict-Transport-Security	Enforces HTTPS (HSTS)
Referrer-Policy: no-referrer	Referrer header privacy leaks



Source Code Blocking Middleware

Blocks direct access to server.js, database.js, .env, package.json, .git, /data/ → returns 403 Forbidden.

✓ ACTIVE



Gmail-Only Registration

Only @gmail.com addresses accepted. Validated both frontend & backend.

✓ ACTIVE



OTP Email Verification

Every new account requires 6-digit OTP. 10-min expiry, 5 max attempts.

✓ ACTIVE



11. Deployment & Hosting

SECTION 11



HOSTING

Render.com

Free Web Service. Auto-deploy from GitHub. HTTPS included.



DATABASE

MongoDB Atlas

Free M0 cluster (512MB). Cloud-hosted, always online.

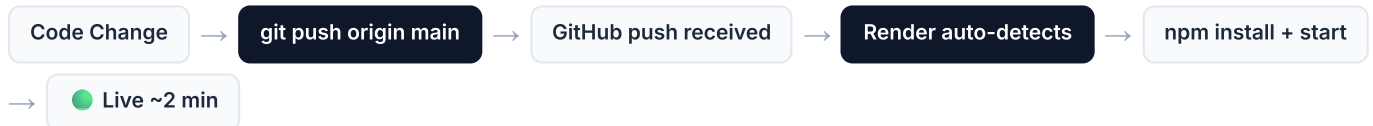


SOURCE CONTROL

GitHub

UrQtSketch/Daily-Grind-Tracker. Auto-deploys on push.

Deployment Flow



Environment Variables (set in Render Dashboard)

Variable	Required For
MONGODB_URI	MongoDB cloud connection. Without it → local JSON fallback used.
SMTP_USER	Gmail sender: support.dailygrind@gmail.com
SMTP_PASS	Gmail App Password (16-char) for SMTP authentication
ADMIN_EMAIL	Additional owner email for admin server-level bypass
ADMIN_KEY	Extra x-admin-key value for API access
PORT	Auto-set by Render (3000 for local dev)



Render Free Tier Cold Start:

Server sleeps after 15 min of no traffic. First visitor waits 30–50 sec. Fix: UptimeRobot.com → ping /api/health every 5 min (free) → server stays awake 24/7.

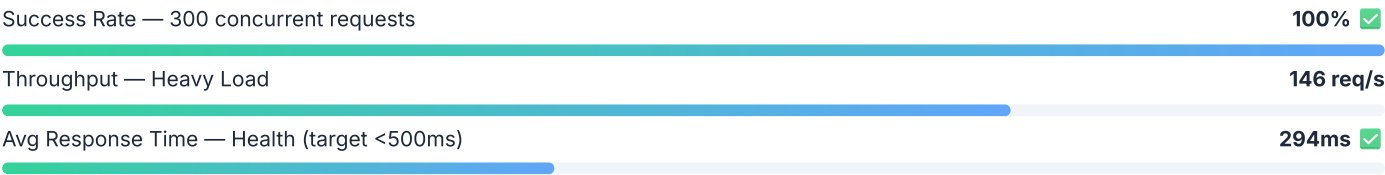


12. Load Test Results

SECTION 12

Tested **September 17, 2026** on live production: <https://daily-grind-tracker.onrender.com>

Test	Requests	Concurrent	Success Rate	Avg Response	Throughput
Light Load	100	20	100%	19ms	54 req/s
Heavy Load	300	50	100%	316ms	146 req/s
Health API	150	30	100%	294ms	94 req/s
Leaderboard API	100	20	100%	1,289ms	14 req/s



✓

ZERO DOWNTIME

300 simultaneous requests — zero failures or timeouts

✓

RATE LIMITING WORKS

After 300 req/10min per IP → 429 Too Many Requests

✓

PRODUCTION READY

Handles expected launch load comfortably



13. Key Credentials & Configuration

SECTION 13



CONFIDENTIAL — Do Not Share Publicly.
Never post these on GitHub, Discord, or any public platform.

URLs & Links

Live Production URL	https://daily-grind-tracker.onrender.com
GitHub Repository	https://github.com/UrQtSketch/Daily-Grind-Tracker
Health Check	https://daily-grind-tracker.onrender.com/api/health
Admin Panel	https://daily-grind-tracker.onrender.com/#admin

Admin Access

Admin PIN 1	grind751
Admin PIN 2	admin2026

Owner Email 1	deepak222@gmail.com
Owner Email 2	support.dailygrind@gmail.com

Hosting Dashboards & Version

Render Dashboard	https://dashboard.render.com
MongoDB Atlas	https://cloud.mongodb.com
App Version	v2.8.1 · styles.css?v=2.8 · app.js?v=2.8
Node.js Minimum	v18.0.0
Email Reminders	Every 30-min check · Sends at 8:00 PM IST only



14. Git History & Local Setup

SECTION 14

Recent Commits (Latest First)

Commit	Description	Key Changes
1cca557	security: helmet + rate-limit DDoS protection	Helmet headers, 3-layer rate limiting added & tested
5806670	feat: Logout Admin button, fix admin bypass	Logout button, removed isAutoAdmin bypass
53afb7a	feat: support inbox in admin, remove PIN hint	Support Messages tab, Gmail reply button
5db1a8c	remove: Basic/Classic Mode toggle from topbar	Cleaned topbar UI
9cc1562	fix: black screen — sidebar-backdrop bug	Critical layout fix
f76aa93	feat: sidebar collapse 3-dots button	Desktop collapse, mobile overlay, 768px breakpoint

How to Run Locally

```
# 1. Clone
git clone https://github.com/UrQtSketch/Daily-Grind-Tracker.git
cd Daily-Grind-Tracker

# 2. Install
npm install

# 3. Start (uses local JSON fallback — no MongoDB needed)
npm start

# Open: http://localhost:3000

# 4. Optional .env for full features:
MONGODB_URI=mongodb+srv://...
SMTP_USER=support.dailygrind@gmail.com
SMTP_PASS=your-gmail-app-password
```

Future Roadmap

PERFORMANCE

Redis cache for leaderboard · CDN for static assets · MongoDB query indexes

PUSH NOTIFICATIONS

PWA Service Worker · Browser notification API · SMS via Twilio

PWA / MOBILE APP

Add to Home Screen manifest · Offline mode · App icons

ADMIN ANALYTICS

User activity graphs · Trophy trend charts · Growth dashboard

Daily Grind Tracker · v2.8.1 · Built by Sketch (UrQtSketch) · September 17, 2026

GitHub: <https://github.com/UrQtSketch/Daily-Grind-Tracker> · Live: <https://daily-grind-tracker.onrender.com>

CONFIDENTIAL — For authorized personnel only